

Generalized reflection principle for random walks

Tarasov Alexander

based on joint works with Vitali Wachtel and Denis Denisov

Hildesheim University

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Classical reflection principle

Consider a random walk S_n and let τ be the first time when S_n becomes non-positive, that is,

$$\tau := \inf_n \{n \geq 1 : S_n \leq 0\}.$$

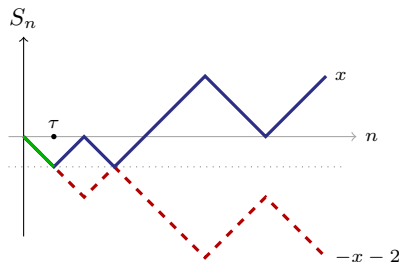
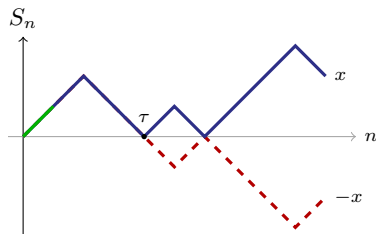
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The classical reflection principle for a simple random walk says

$$\mathbb{P}(S_n \geq x, \tau < n) = \frac{1}{2} \mathbb{P}(1 + S_{n-1} \leq -x) + \frac{1}{2} \mathbb{P}(-1 + S_{n-1} \leq -x - 2).$$



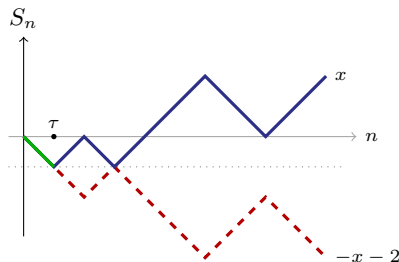
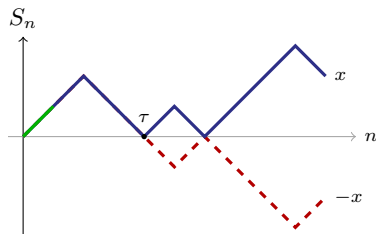
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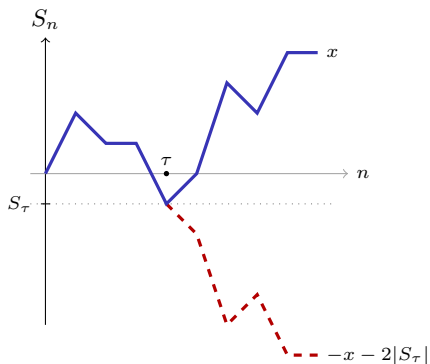
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We treat the first step separately because of the overshoot, which shifts the reflection from $-x$ to $-x - 2$.

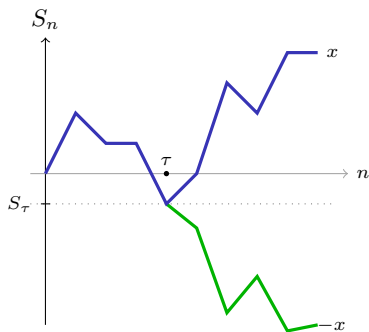
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For a general random walk we have an overshoot $|S_\tau|$ whose distribution can be determined exactly only in very special cases.



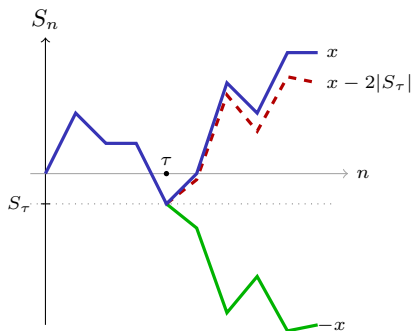
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Generalized reflection principle

Consider a discrete symmetric random walk, for $x \geq 0$ we have

$$\mathbb{P}(S_n \geq x, \tau > n) = \mathbb{P}(S_n \geq x) - \mathbb{P}(S_n \geq x, \tau \leq n).$$

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Let us reflect the walk but keep the condition on τ :

$$0 = \mathbb{P}(S_n \leq -x, \tau > n) = \mathbb{P}(S_n \leq -x) - \mathbb{P}(S_n \leq -x, \tau \leq n).$$

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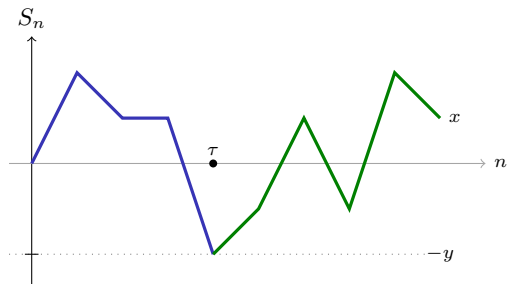
Taking the difference we obtain

$$\mathbb{P}(S_n \geq x, \tau > n) = \mathbb{P}(S_n \leq -x, \tau \leq n) - \mathbb{P}(S_n \geq x, \tau \leq n).$$

Generalized reflection principle

One can use the full probability formula for the stopping time τ :

$$\mathbb{P}(S_n \geq x, \tau \leq n) = \sum_{k=1}^{n-1} \sum_{y \geq 0} \mathbb{P}(S_k = -y, \tau = k) \mathbb{P}(S_{n-k} \geq x + y)$$



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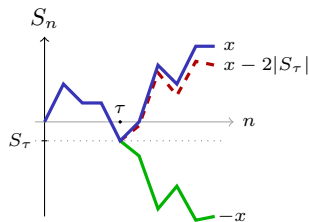
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Berry-Esseen inequality

Theorem

Assume that $\mathbb{E}X_1 = 0$, $\mathbb{E}X_1^2 = 1$, and $\mathbb{E}|X_1|^3 < \infty$. Then there exists an absolute constant A such that

$$\sup_{x \geq 0} \left| \mathbb{P} \left(\frac{S_n}{\sigma \sqrt{n}} \geq x \mid \tau > n \right) - e^{-x^2/2} \right| \leq A \frac{(\mathbb{E}|X_1|^3)^3}{\sqrt{n}}.$$

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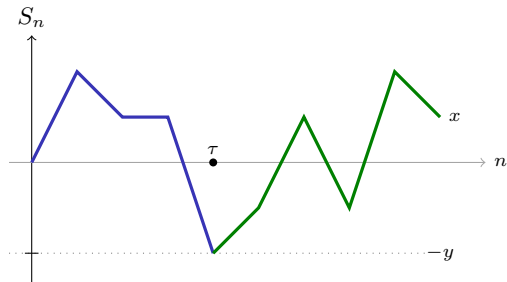
$$\begin{aligned} \left| \mathbb{P}(y + S_n \geq x, \tau_y > n) - \mathbb{P}(y + B_n \geq x, \tau_y^{bm} > n) - \frac{2}{\sqrt{2\pi n}} e^{-\frac{x^2}{2n}} \mathbb{E}|y + S_{\tau_y}| \right| \\ \leq A_1 \frac{(\mathbb{E}|X_1|^3)^3 \mathbb{E}|S_{\tau_y}|}{\sqrt{n}(y + \sqrt{n})}. \end{aligned}$$

where B_t is a Brownian motion and τ_y^{bm} is a stopping time for a Brownian motion started at y .

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There are two shortcomings:

- ▶ We expect F_ν to be polynomials;
- ▶ The coefficients F_ν practically are not computable.

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Or equivalently

$$P_\nu(u) - R_\nu(u) = -Q_\nu(u).$$

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Thus, the theorem takes a much cleaner form:

Theorem

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$$\mathbb{P}(S_n = x, \tau > n) = e^{-u^2/2} \sum_{\nu=1}^{r+2} \frac{-2Q_\nu(u)}{n^{\nu/2}} + O\left(n^{-\frac{r+2}{2}}\right),$$

where $x \sim u\sqrt{n} > 0$ and Q_ν are computable polynomials.

Random walk in a cone

A natural generalization of a random walk conditioned to stay positive is a random walk conditioned to stay in a cone. For a d -dimensional random walk S_n and a cone K we define

$$\tau = \inf_n \{n \geq 1 \mid S_n \notin K\}.$$

It is known that for a convex cone in $\mathbb{R}^d$ under a moment assumption it holds

$$\mathbb{P}(\tau > n) = \frac{C}{n^{p/2}} (1 + o(1)),$$

where constants C and p depend on the cone's geometry. Estimates sharper than $o(1)$ are known only in a few special cases.

Reflection in $\mathbb{R}^2$

The first step of our method can be carried out

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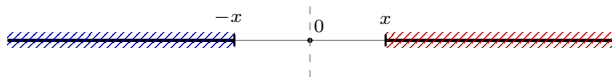
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The question is: what is the counterpart of the reflected set? Let us examine the one-dimensional case, where we compare

$$\mathbb{P}(S_n \in (-\infty, x], \tau > n) \quad \text{and} \quad \mathbb{P}(S_n \in [x, \infty), \tau > n).$$



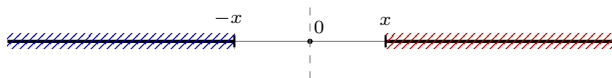
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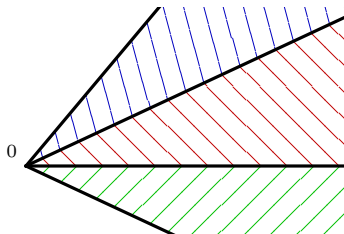


Then symmetry (or asymptotic symmetry) of S_n gives us a cancellation in the difference

$$\mathbb{P}(S_n \in [x, \infty)) - \mathbb{P}(S_n \in (-\infty, x]).$$

Reflection in $\mathbb{R}^2$

For a two-dimensional cone we have at least two natural reflections:

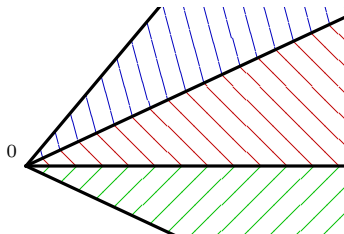


Let S_n has a unit covariance matrix, then in main asymptotic all probabilities coincide

$$\mathbb{P}(S_n \in \textcolor{red}{K}) \sim \mathbb{P}(S_n \in \textcolor{blue}{K}^{\text{left}}) \sim \mathbb{P}(S_n \in \textcolor{green}{K}^{\text{right}}),$$

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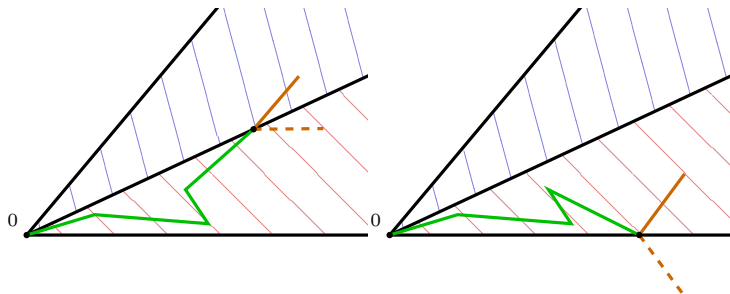
and so one can write

$$\begin{aligned}\mathbb{P}(S_n \in K, \tau > n) &= \mathbb{P}(S_n \in K, \tau > n) - \mathbb{P}(S_n \in K^{\text{left}}, \tau > n) \\ &= \mathbb{P}(S_n \in K) - \mathbb{P}(S_n \in K^{\text{left}}) \\ &\quad + \mathbb{P}(S_n \in K^{\text{left}}, \tau \leq n) - \mathbb{P}(S_n \in K, \tau \leq n).\end{aligned}$$

Reflection in $\mathbb{R}^2$

Let us look precisely at the last difference:

$$\mathbb{P}(S_n \in K^{\text{left}}, \tau \leq n) - \mathbb{P}(S_n \in K, \tau \leq n).$$



A different way to reflect

Let $\mathbb{P}_{\geq 0}$ be the probability conditioned on the second coordinate being non-negative. In particular, the random walk does not leave the upper half-plane. Then

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